

Exercise: Build an ER Diagram for Company HR Database

The company has

Employees

Departments

Projects

Managers

Rules

- Every department has exactly one manager.
- A manager is also an employee.
- Employees belong to one department.
- Employees can work on multiple projects.
- Each project has a budget.
- Store how many hours each employee spends on each project.

This one requires thinking about:

- Recursive relationships (Manager is an Employee)
- One-to-many relationships
- Many-to-many relationships
- Bridge tables

SQLite ERD to Table Exercise: Company Database

Step 1: Identify Tables

Create tables for:

- 1.
- 2.
- 3.
- 4.

Step 2: Identify Columns

Employee Table:

-
-
-
-

Department Table:

-
-
-

Project Table:

-
-
-

Employee_Project Table:

-

-
-

Step 3: Define Primary Keys

Employee Primary Key: _____

Department Primary Key: _____

Project Primary Key: _____

Employee_Project Primary Key: _____

Step 4: Define Relationships

Department → Employee

Relationship: _____

Foreign Key: _____

Employee → Project

Relationship: _____

Bridge Table: _____

Step 5: Add Foreign Keys

Employee Table:

Foreign Key: _____

Employee_Project Table:

Foreign Key 1: _____

Foreign Key 2: _____

Step 6: Draw ERD

Include:

- Tables
- Primary Keys
- Foreign Keys
- Relationships

Step 7: Create SQLite Database

Database Name:

_____.db

Step 8: Create SQL Tables

Create tables in order:

- 1.
- 2.
- 3.
- 4.

Step 9: Insert Data

Insert:

- 3 Departments
- 5 Employees

- 3 Projects
- Employee Project assignments

```
CREATE TABLE Employee (  
    Employee_ID INTEGER PRIMARY KEY,  
    First_Name TEXT NOT NULL,  
    Last_Name TEXT NOT NULL,  
    Email TEXT UNIQUE NOT NULL,  
    Phone_Number TEXT UNIQUE,  
    Job_Title TEXT NOT NULL,  
    Salary REAL CHECK(Salary > 0),  
    Hire_Date DATE DEFAULT CURRENT_DATE,  
    Employment_Status TEXT DEFAULT 'Active'  
        CHECK(Employment_Status IN ('Active', 'Inactive')),  
    Department_ID INTEGER,  
    Manager_ID INTEGER,  
  
    FOREIGN KEY (Department_ID)  
        REFERENCES Department(Department_ID),  
  
    FOREIGN KEY (Manager_ID)  
        REFERENCES Employee(Employee_ID)
```

);